

Does Video Count Drive Creator Success?

YouTube Creator Analytics · UIUC MSBA · 2026

Yun-Chen (Sunny) Lee

75

Leading
Creators

3

Content Categories

YouTube
Data API v3

Project Overview

Common Assumption

*"The more frequently you upload,
the faster your channel grows."*

H_0 (Null):

Upload frequency has NO significant effect on subscriber count.

H_1 (Alternative):

Upload frequency positively predicts creator success.

01

Data Collection

YouTube Data API v3 · 75 leading creators collected via API search across 3 categories

02

Data Preparation

Merged CSVs · Engineered subs_per_video & views_per_video

03

Exploratory Analysis

Correlation matrix · Log-scale scatter · Category medians

04

Statistical Testing

OLS Regression + Kruskal-Wallis non-parametric test

05

Tableau Dashboard

3 interactive dashboards covering landscape, performance & trends

Dataset & Sample Design



25 Creators

Entertainment



25 Creators

Education



25 Creators

Science &
Technology

Variables Collected

Variable	Type	Description	Role
subscriber_count	Continuous	Total channel subscribers	Outcome Variable
video_count	Continuous	Total videos published	Predictor Variable
view_count	Continuous	Total channel views	Supplementary
category	Categorical	Entertainment / Education / Science & Tech	Group Variable

Statistical Results — OLS Regression

0.025

R^2

Explains only 2.5% of subscriber variance

0.156

Coefficient

Weak positive trend (more videos → slightly more subs)

0.175

p-value

NOT significant — fail to reject H_0 ($p > 0.05$)

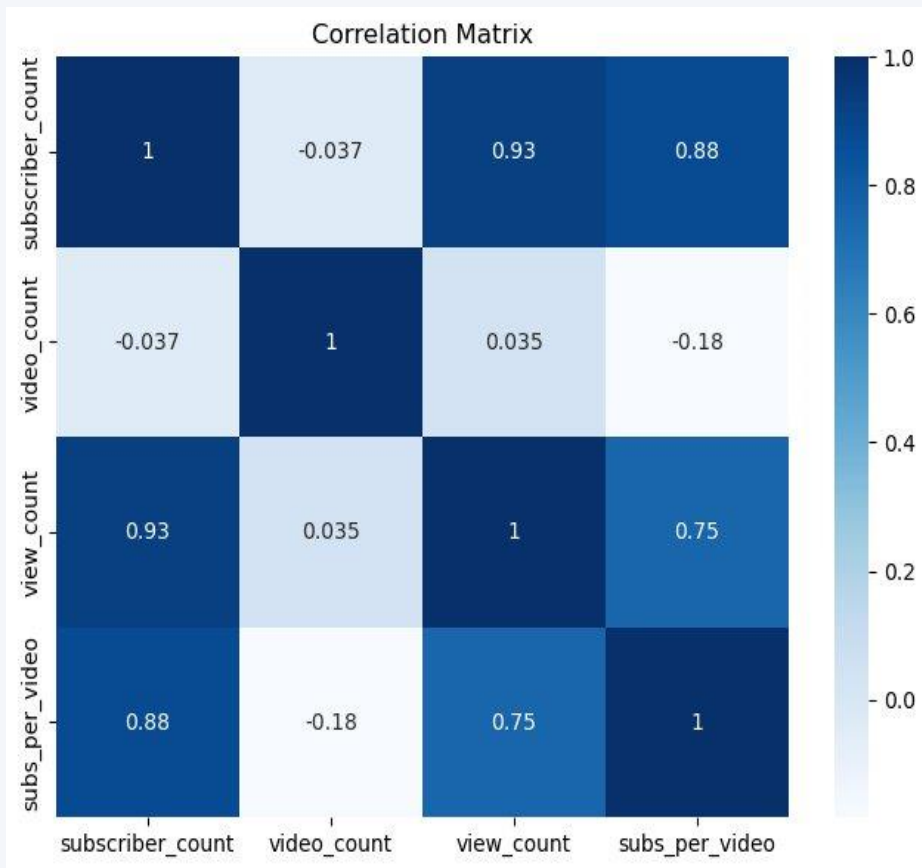
OLS Regression Results

```
=====
Dep. Variable:    subscriber_count    R-squared:                0.025
Model:            OLS                Adj. R-squared:           0.012
Method:           Least Squares       F-statistic:              1.872
Date:             Mon, 08 Jun 2026    Prob (F-statistic):       0.175
Time:             11:50:33            Log-Likelihood:           -118.32
No. Observations: 75                 AIC:                     240.6
Df Residuals:     73                 BIC:                     245.3
Df Model:          1
Covariance Type:  nonrobust
=====
```

```
=====
              coef    std err          t      P>|t|      [0.025    0.975]
-----
const         15.0130     0.811     18.510     0.000     13.396     16.629
video_count    0.1557     0.114      1.368     0.175     -0.071     0.383
=====
```

```
=====
Omnibus:                 3.669    Durbin-Watson:           0.901
Prob(Omnibus):            0.160    Jarque-Bera (JB):        4.008
Skew:                     0.017    Prob(JB):                0.135
Kurtosis:                 4.132    Cond. No.                 43.0
=====
```

Correlation Matrix



Key Correlations

subscriber ↔ video_count

$r = -0.037$

Near-zero — upload volume does NOT drive subs

subscriber ↔ view_count

$r = 0.93$

Very strong — engagement drives channel size

subs_per_video ↔ video_count

$r = -0.18$

More videos → slightly lower efficiency

Category Comparison — Kruskal-Wallis Test

Kruskal-Wallis Test → Statistic: 28.74 | p-value: 5.72×10^{-7} | Result: Highly Significant ($p < 0.001$)

Entertainment



Most Efficient

Avg. Subscribers

44.5M

Avg. Videos

1,800

Median Subs / Video

27,979

Education



2nd

Avg. Subscribers

14.7M

Avg. Videos

2,255

Median Subs / Video

11,274

Science & Tech



3rd

Avg. Subscribers

6.9M

Avg. Videos

2,955

Median Subs / Video

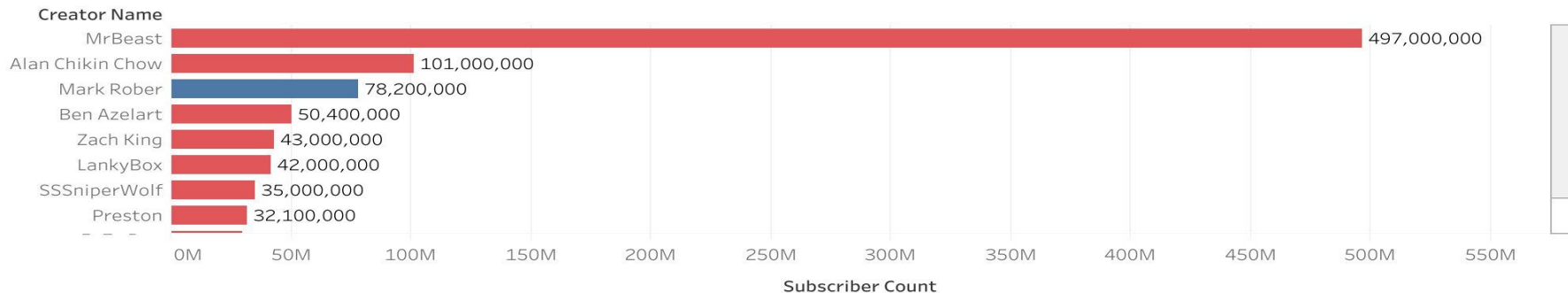
3,120

Median subs/video used to reduce skew from outliers (MrBeast: 504K, Mark Rober: 303K subs/video).

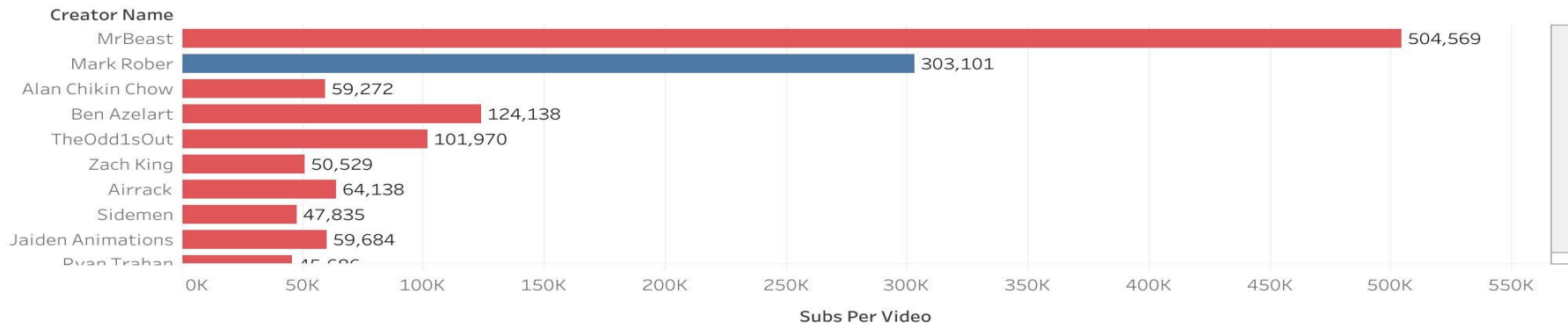
Results remain consistent when using median efficiency, reducing the impact of outliers such as MrBeast and Mark Rober.

Creator Performance — Top Rankings

Top 10 Creators by Subscribers



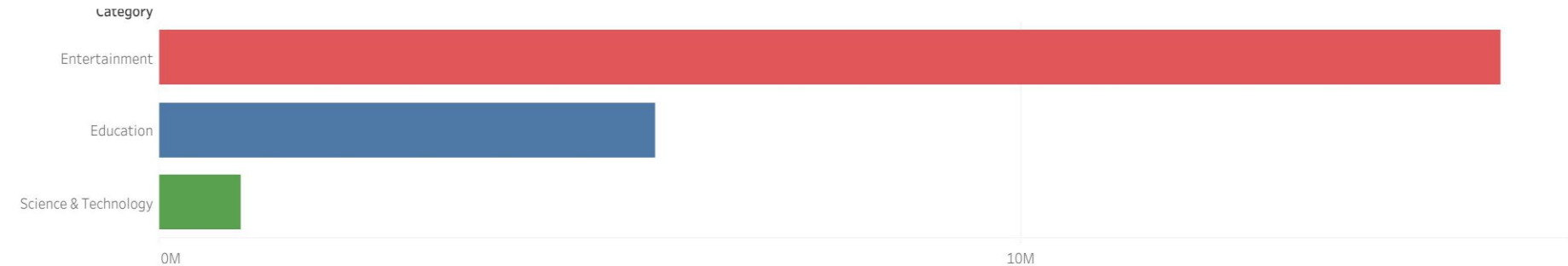
Top 10 Most Efficient Creators



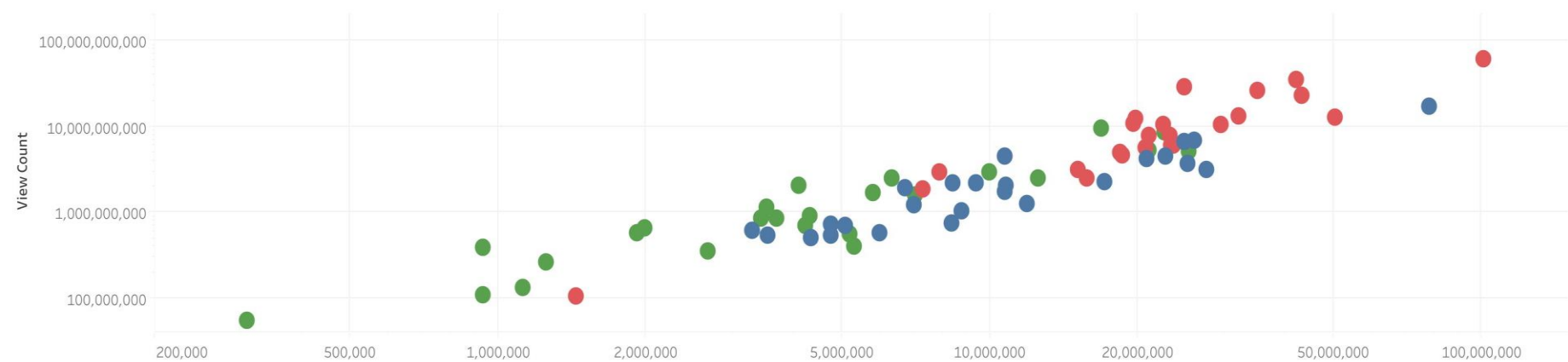
YouTube Creator Landscape — Category Overview

Category All

Category Summary

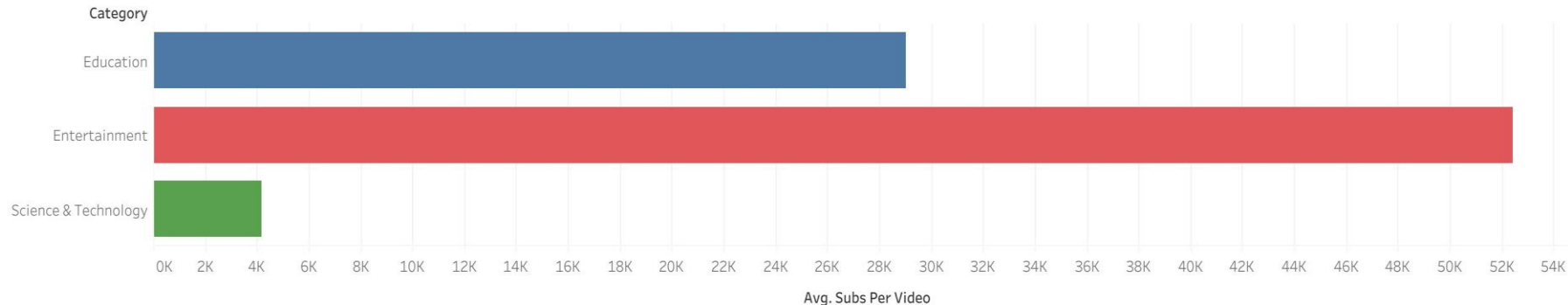


Subscriber vs Views

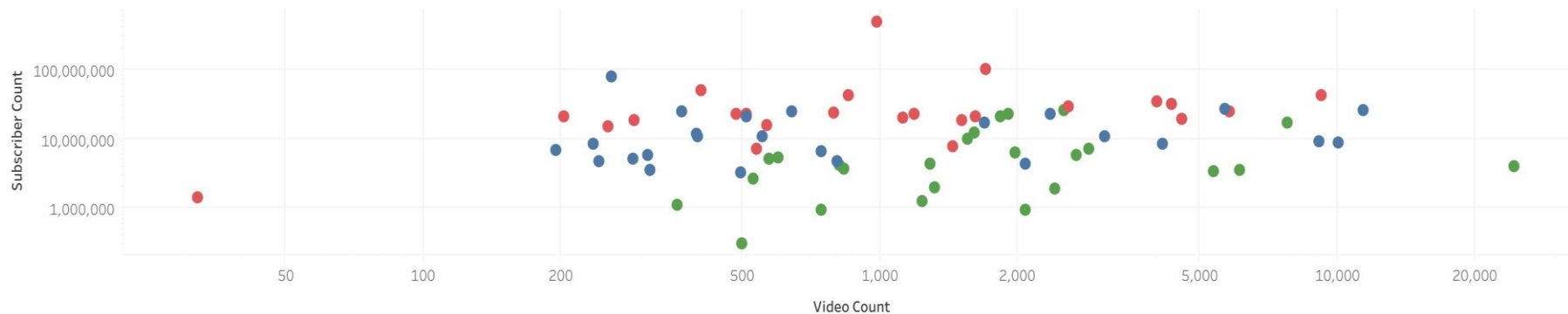


Content Efficiency & Video Count vs. Subscribers

Average Content Efficiency by Category



Video Count vs Subscriber Count



Key Findings

1

Video Count Does NOT Drive Growth

$R^2 = 0.025$, $p = 0.175$ — fail to reject H_0 .
Video count explains only 2.5% of variation in subscriber count.

2

Near-Zero Correlation Confirmed

Correlation between video_count & subscriber_count is -0.037. Higher upload volume links to lower per-video efficiency ($r = -0.18$).

3

Engagement — Not Volume — Drives Growth

subscriber_count $\leftrightarrow$ view_count
correlation: $r = 0.93$. Audience engagement is the primary channel growth driver.

4

Entertainment is 12.6× More Efficient

52,416 subs/video vs 4,150 for Science & Tech. Kruskal-Wallis confirms category differences are highly significant ($p = 5.72 \times 10^{-7}$).

5

Quality Outliers Beat High-Volume Creators

Mark Rober: 78M subs, only 258 videos $\rightarrow$ 303K subs/video. MrBeast, Kurzgesagt & TheOdd1sOut show the same pattern.

Conclusion

Upload frequency alone is NOT a meaningful predictor of YouTube creator success.

- Video count explains only 2.5% of variation in subscriber count ($R^2 = 0.025$)
- The correlation between upload volume and subscribers is effectively zero ($r = -0.037$)
- Content category, engagement quality, and audience fit are far more important than posting frequency
- Entertainment creators achieve 12.6× higher subscriber efficiency than Science & Technology creators — with fewer videos

Strategic Recommendation: Invest in content quality and audience alignment — not upload frequency.